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Negative poisson's ratio materials for intraocular lenses

Abstract

An intraocular lens includes a substantially circular lens element formed of a transparent material and one or more haptics extending outwardly from an outer edge of the lens element. The one or more haptics are formed of a polymer foam material having a negative Poisson's ratio (NPR) and are configured to couple the intraocular lens to an eye of a patient. The lens includes an inner region having a first index of refraction and an outer region disposed circumferentially surrounding the inner region, the outer region having a second index of refraction different from the first index of refraction.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
4834750	12/1988	Gupta	N/A	N/A
5628798	12/1996	Eggleston et al.	N/A	N/A
6013101	12/1999	Israel	N/A	N/A
8216310	12/2011	Hu et al.	N/A	N/A
12042375	12/2023	Park	N/A	A61F 2/1453
2005/0021139	12/2004	Shadduck	N/A	N/A
2005/0277864	12/2004	Haffner et al.	N/A	N/A
2006/0116765	12/2005	Blake et al.	N/A	N/A
2023/0149153	12/2022	Park	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO 199007575	12/1989	WO	N/A

OTHER PUBLICATIONS

Bhushan et al., “Adhesion and friction properties of polymers in microfluidic device,” Nanotechnology, Jan. 11, 2005, 16:467-478. cited by applicant
PCT International Search Report and Written Opinion in International Appln. No. PCT/US22/49948, dated Feb. 22, 2023, 9 pages. cited by applicant
Song et al., “Controllable fabrication of porous PLGA/PCL bilayer membrane for GTR using supercritical carbon dioxide foaming,” Applied Surface Science, Apr. 2018, 472:82-92. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application of and claims the benefit of priority to U.S. application Ser. No. 17/527,830, filed on Nov. 16, 2021, the contents of which are hereby incorporated by reference.

BACKGROUND

(1) The present disclosure relates generally to materials for intraocular lenses. An intraocular lens is a device that is surgically placed in an eye, e.g., to replace or supplement the focusing power of a natural lens.

SUMMARY

(2) We describe here intraocular lenses that include materials having a negative Poisson's ratio (“NPR materials”).

(3) In an aspect, the intraocular lens includes a substantially circular lens element formed of a transparent material and one or more haptics extending outwardly from an outer edge of the lens element, the one or more haptics formed of a polymer foam material having a negative Poisson's ratio (NPR), in which the one or more haptics are configured to couple the intraocular lens to an eye of a patient. The lens includes an inner region having a first index of refraction and an outer region disposed circumferentially surrounding the inner region, the outer region having a second index of refraction different from the first index of refraction.

(4) Embodiments of the intraocular lens can include one or any combination of two or more of the following features.

(5) The outer region of the intraocular lens element is formed of a polymer foam NPR material. The inner region of the intraocular lens element is formed of acrylic (e.g., polymethylmethacrylate), silicone, or hydrogel.

(6) The intraocular lens includes two haptics disposed at diametrically opposed positions around a circumference of the lens element. The intraocular lens element includes four haptics disposed at evenly spaced intervals around a circumference of the lens element. The haptics include one or more loop haptics. The haptics include one or more plate loop haptics. The haptics include one or more solid plate haptics. The haptics include one or more T-shaped haptics. The width of each haptic is less than the diameter of the lens element. The width of each haptic is substantially equal to a diameter of the lens element. One or more haptics include a hinge attaching the haptic to the lens element. The hinge attaching the haptic to the lens element includes grooves in the haptic. The lens element is convex. The polymer foam material of the lens element is composed of a cellular structure having a characteristic dimension of between 0.1 μm and 3 μm. The polymer foam material includes a foam of acrylic (e.g., polymethylmethacrylate), silicone, or hydrogel.

(7) In an aspect, a method of implanting an intraocular lens in an eye of a patient includes creating an incision in a cornea of the eye of the patient, inserting the intraocular lens into the eye of the patient through the incision in the cornea, and securing the intraocular lens in the eye such that a substantially circular, transparent lens element of the intraocular lens is disposed along an optical axis of the eye and one or more haptics of the intraocular lens are in contact with tissue of the eye, in which the one or more haptics are formed of a polymer foam material having a negative Poisson's ratio.

(8) Embodiments of the method of implanting an intraocular lens in an eye of a patient can include one or any combination of two or more of the following features. The existing lens is removed from the eye of the patient prior to inserting the intraocular lens. The intraocular lens is inserted into a posterior chamber of the eye. The intraocular lens is inserted into an anterior chamber of the eye.

(9) Other implementations are within the scope of the claims.

Description

DESCRIPTION OF DRAWINGS

(1) FIG. **1** is an illustration of components of the human eye.

(2) FIG. **2** is one embodiment of two-loop intraocular lens.

(3) FIG. **3** is an illustration of materials with negative and positive Poisson's ratios.
(4) FIG. **4** is an illustration of balls with negative and positive Poisson's ratios.
(5) FIGS. **5**A and **5**B are plots of diameter versus time.
(6) FIG. **6** is an illustration of composite materials.
(7) FIG. **7** is an illustration of a material with a positive Poisson's ratio and a composite material.
(8) FIG. **8** is a diagram of a method of making an NPR material.
(9) FIGS. **9**A-**9**F are illustrations of intraocular lenses.
(10) FIG. **10** is a flow chart.

DETAILED DESCRIPTION

(11) This disclosure describes intraocular lenses (IOLs) formed in part from materials having a negative Poisson's ratio (“NPR materials”). An intraocular lens is a device that is surgically placed in an eye, e.g., to replace or supplement the focusing power of a natural lens, e.g., if the natural lens becomes clouded by cataracts, damaged, or lacks sufficient focusing power. The inclusion of NPR materials in intraocular lenses can produce intraocular lenses which are lighter, more comfortable and easier to implant than lenses made with conventional materials (e.g., materials having a positive Poisson's ratio (“PPR materials”). Surgical procedures used to place an IOL within the eye of a patient are also described in this disclosure.

(12) FIG. **1** shows a schematic of a human eye **100**. A sclera **102** is the white outermost layer of the eye that covers and maintains the shape of the eye. A cornea **104** is on the front, exterior of the eye and is made of transparent tissue that acts to refract light entering a pupil **106**. An anterior chamber **108** is a fluid-filled cavity in the front part of the eye between the cornea **104** and an iris **110**. The iris **110** controls the amount of light that enters the eye by controlling the diameter of the pupil **106**. A posterior chamber **112** is a fluid-filled cavity between a lens **114** and the iris **110**. A capsular bag **116** is a structure that holds the lens **114** in a central position within the eye. Attached to the lens **114** are small suspensory ligaments (zonules) **118** extending from the inner wall of the eye that connect to a ciliary body **120**. The ciliary body **120** is the middle layer of the wall of the eye and includes a ring shaped muscle. A ciliary sulcus **122** is a small space between the posterior surface of the iris **110** and the anterior surface the ciliary body **120**. The lens **114** is elastic and can change its shape when pulled on by the ciliary body muscles **120**.

(13) Approximately two-thirds of the focusing power of the eye is achieved by the cornea **104**. The remaining approximately one-third of the focusing power is achieved by the lens **114**. Unlike the cornea **104**, which remains static as it refracts light, the action of the ciliary body muscles **120** allows the lens **114** to dynamically change in order to vary the distance at which the eye can focus an image: the lens **114** can become thicker to focus on nearby objects and thinner to focus on distant objects. After being refracted through the lens, the fully focused light impinges on a retina **124**, is converted to neural signals, and is transmitted to the brain for image formation by an optic nerve **126**.

(14) As humans age, the lens **114** can stiffen and lose the flexibility which allows the eye to focus on objects over the full range of distances. Proteins in the lens can break down over time, leaving the lens cloudy and impeding its ability to transmit and focus light. These protein deposits, called cataracts, may become so severe that the removal and replacement of the lens becomes appropriate. An intraocular lens is a device that is surgically placed in the eye, e.g., to replace or supplement the focusing power of the natural lens **114**, e.g., if the natural lens becomes clouded by cataracts, damaged, or lacks sufficient focusing power. An IOL may be implanted into the eye following removal of the natural lens **114**, or may be placed over the existing natural lens **114** to change the focusing power of the eye. Examples of conditions that may be treated by the replacement of the lens of the eye with an intraocular lens or the insertion of an IOL over the lens include hyperopia, myopia, presbyopia, astigmatism, eye trauma, and genetic eye conditions.

(15) The anterior and posterior surfaces of an intraocular lens (IOL) each has a radius of curvature designed to refract light. Each of the two surfaces can be flat (with a radius of curvature equal to

infinity) or convex (with a radius of curvature greater than zero and less than infinity). A convex surface can have a radius of curvature that is set depending on the function of the lens. In some lenses, the radius of curvature of the anterior surface differs from the radius of curvature of the posterior surface of the lens. Biconvex lenses have two convex surfaces with different radii of curvature for the surfaces. Equiconvex lenses have two convex surfaces with the same radius of curvature for both surfaces. Planoconvex lenses have one flat surface and one convex surface. The aperture of these refracting surfaces defines the optical zone of the IOL.

(16) Intraocular lenses include four general functional types: monofocal, toric, presbyopic-correction, and phakic. Monofocal IOLs can be used to correct vision at a single distance range, and these lenses can be configured to correct either nearsightedness or farsightedness. Toric IOLs can be used for patients who have corneal astigmatism. Toric IOLs have markers on the peripheral parts of the lens that enable the surgeon to see the orientation of the astigmatism correction in the lens. Once the toric IOL is implanted in the eye, the surgeon then rotates the lens so the astigmatism correction is properly aligned for best results. Monofocal or toric IOLs generally correct only one type of visual deficiency. Presbyopic-correcting IOLs can be used to correct multiple vision deficiencies and include the following general functional types: multifocal IOLs which provide multiple zones of lens power that produce more than one focal point; bifocal diffractive IOLs which create two distinct images at near and far distant ranges; trifocal diffractive IOLs which improve intermediate vision compared to bifocal IOLs by providing a third range of focus; refractive IOLs which create multiple focal points that allow viewing at all distances; extended depth of focus IOLs which are designed to give an elongated focus of vision without compromising distance visual acuity; accommodative IOLs which simulates the natural accommodative process in the eye by changing power in response to ciliary muscle contraction. A phakic IOL is placed over the existing natural lens and is used to change the eye's focusing power.

(17) A two-loop embodiment of an intraocular lens **200** is shown in FIG. **2**. A central, substantially circular portion of the intraocular lens **200** is called an optic **202**. A region of the optic **202** which focuses the light is called an optical zone **204**. The optical zone **204** is a rounded (e.g., generally circular) region in an interior of the optic **202**. The optical zone **204** has a first index of refraction that, when the IOL **200** is implanted in an eye, facilitates focusing of light. The shape, curvature, or both of the optical zone **204** can facilitate focusing of light. The optic **202** includes a circumferential optic **206** concentrically surrounding the outer circumference of the optical zone **204**. The circumferential optic **206** can be made of a different material than the optical zone and can have a second index of refraction different from the first index of refraction.

(18) Haptics **208** extend outwardly from the outer edge of the optic **202**. The haptics are flexible extensions that help to stabilize and center the lens within the eye after implantation. The haptics can be attached to the optic **202** at the circumferential optic **206**. The IOL depicted in FIG. **2** is a two-loop intraocular lens **200** with two loop haptics **208** at diametrically opposed positions around the circumferential optic **206**. Each haptic **208** decreases in width with increasing distance from the optic, from an inner width **210** to an outer width **212**, with the outer width **212** less than the inner width **210**. Both the inner width **210** and the outer width **212** are significantly less than a diameter of the optic **202**.

(19) The optical zone **204** can be formed from a transparent material, e.g., a transparent polymer material, having a positive Poisson's ratio (PPR) with a first index of refraction. The material used to form the optical zone can include acrylic (e.g., polymethyl methacrylate (PMMA)), polycarbonate, silicone, or hydrogel, or a combination thereof. One or more of the haptics **208**, the circumferential optic **206**, or all of them can be formed of a material having a negative Poisson's ration (NPR), e.g., an NPR foam material, such as an NPR polymer foam material or an NPR polymer sponge material. In some examples, the circumferential optic **206** and both haptics **208** are made from the same NPR material. In some examples, the circumferential optic **206** is made from one NPR material, and the haptics **208** are made from a different NPR material. In some examples,

the circumferential optic **206** and each haptic **208** is each made from a different NPR material.
(20) The IOL **200** can be fabricated as a single piece or a multi-piece lens. All components of a single piece lens are manufactured from the same material. A multi-piece lens can have the optical zone **204** of the optic **202**, the circumferential optic **206** of the optic, and the haptics **208** fabricated from different materials. The circumferential optic **206**, which provides a point of attachment for the haptics **208**, can be made of a different material than either the haptics **208** or the optical zone **204** of the intraocular lens. The disclosure describes the use of a negative Poisson's ratio (NPR) material for the haptics **208**, the circumferential optic **206**, or both, in an intraocular lens **200**.
(21) Properties that may be important for materials used for intraocular lenses include biocompatibility, infection resistance, clarity, refractive index, durability, flexibility, UV filtration, or any combination of these properties. Materials generally undergo quality control to ensure they meet criteria appropriate to be used for IOLs. Although polymethylmethacrylate (PMMA) was used extensively for early versions of IOLs and is still in use today, other polymer materials also have been implemented in IOLs. Examples of materials that are used to fabricate intraocular lenses include acrylics, silicone, polycarbonate, or combinations thereof. Acrylic materials can be rigid (e.g., PMMA) or flexible. The acrylic materials can be either hydrophobic or hydrophilic. Hydrogels, which include hydrophilic polymers of acrylic and silicone, also can be used to fabricate IOLs. The disclosure describes the use of a negative Poisson's ratio (NPR) material for the haptics **208**, the circumferential optic **206** of the optic **202**, or both, in an intraocular lens. NPR materials used in IOLs can be selected to have properties suitable for use in intraocular lenses. For instance, the haptics **208**, the circumferential optic **206**, or both can incorporate NPR materials of acrylics (e.g., PMMA), silicone, polycarbonate, or hydrogels.
(22) An NPR material is a material that has a Poisson's ratio that is less than zero, such that when the material experiences a positive strain along one axis (e.g., when the material is stretched), the strain in the material along the two perpendicular axes is also positive (e.g., the material expands in cross-section). Conversely, when the material experiences a negative strain along one axis (e.g., when the material is compressed), the strain in the material along a perpendicular axis is also negative (e.g., the material compresses along the perpendicular axis). By contrast, a material with a positive Poisson's ratio (a “PPR material”) has a Poisson's ratio that is greater than zero. When a PPR material experiences a positive strain along one axis (e.g., when the material is stretched), the strain in the material along the two perpendicular axes is negative (e.g., the material compresses in cross-section), and vice versa.
(23) Materials with negative and positive Poisson's ratios are illustrated in FIG. **3**, which depicts a hypothetical two-dimensional block of material **300** with length l and width w.
(24) If the hypothetical block of material **300** is a PPR material, when the block of material **300** is compressed along its width w, the material deforms into the shape shown as block **302**. The width w**1** of block **302** is less than the width w of block **300**, and the length l**1** of block **302** is greater than the length l of block **300**: the material compresses along its width and expands along its length.
(25) By contrast, if the hypothetical block of material **300** is an NPR material, when the block of material **300** is compressed along its width w, the material deforms into the shape shown as block **304**. Both the width w**2** and the length l**2** of block **304** are less than the width w and length l, respectively, of block **300**: the material compresses along both its width and its length.
(26) NPR materials intraocular lenses can be foams, such as polymeric foams, ceramic foams, metal foams, or combinations thereof. A foam is a multi-phase composite material in which one phase is gaseous and the one or more other phases are solid (e.g., polymeric, ceramic, or metal). Foams can be closed-cell foams, in which each gaseous cell is sealed by solid material; open-cell foams, in which each cell communicates with the outside atmosphere; or mixed, in which some cells are closed and some cells are open.
(27) An example of an NPR foam structure is a re-entrant structure, which is a foam in which the walls of the cells are concave, e.g., protruding inwards toward the interior of the cells. In a re-

entrant foam, compression applied to opposing walls of a cell will cause the four other, inwardly directed walls of the cell to buckle inward further, causing the material in cross-section to compress, such that a compression occurs in all directions. Similarly, tension applied to opposing walls of a cell will cause the four inwardly directed walls of the cell to unfold, causing the material in cross-section to expand, such that expansion occurs in all directions. NPR foams can have a Poisson's ratio of between −1 and 0, e.g., between −0.8 and 0, e.g., −0.8, −0.7, −0.6, −0.5, −0.4, −0.3, −0.2, or −0.1. NPR foams can have an isotropic Poisson's ratio (e.g., Poisson's ratio is the same in all directions) or an anisotropic Poisson's ratio (e.g., Poisson's ratio when the foam is strained in one direction differs from Poisson's ratio when the foam is strained in a different direction).

(28) An NPR foam can be polydisperse (e.g., the cells of the foam are not all of the same size) and disordered (e.g., the cells of the foam are randomly arranged, as opposed to being arranged in a regular lattice). An NPR foam can have a characteristic dimension (e.g., the size of a representative cell, such as the width of the cell from one wall to the opposing wall) ranging from 0.1 μm to about 3 mm, e.g., about 0.1 μm, about 0.5 μm, about 1 μm, about 10 μm, about 50 μm, about 100 μm, about 500 μm, about 1 mm, about 2 mm, or about 3 mm.

(29) In some examples, NPR foams are produced by transformation of PPR foams to change the structure of the foam into a structure that exhibits a negative Poisson's ratio. In some examples, NPR foams are produced by transformation of nanostructured or microstructured PPR materials, such as nanospheres, microspheres, nanotubes, microtubes, or other nano- or micro-structured materials, into a foam structure that exhibits a negative Poisson's ratio. The transformation of a PPR foam or a nanostructured or microstructured material into an NPR foam can involve thermal treatment (e.g., heating, cooling, or both), application of pressure, or a combination thereof. In some examples, PPR materials, such as PPR foams or nanostructured or microstructured PPR materials, are transformed into NPR materials by chemical processes, e.g., by using glue. In some examples, NPR materials are fabricated using micromachining or lithographic techniques, e.g., by laser micromachining or lithographic patterning of thin layes of material. In some examples, NPR materials are fabricated by additive manufacturing (e.g., three-dimensional (3D) printing) techniques, such as stereolithography, selective laser sintering, or other appropriate additive manufacturing technique.

(30) In an example, a PPR thermoplastic foam, such as an elastomeric silicone film, can be transformed into an NPR foam by compressing the PPR foam, heating the compressed foam to a temperature above its softening point, and cooling the compressed foam. In an example, a PPR foam composed of a ductile metal can be transformed into an NPR foam by uniaxially compressing the PPR foam until the foam yields, followed by uniaxially compression in other directions.

(31) In some examples, NPR foams are produced by transformation of PPR foams to change the structure of the foam into a structure that exhibits a negative Poisson's ratio. In some examples, NPR foams are produced by transformation of nanostructured or microstructured PPR materials, such as nanospheres, microspheres, nanotubes, microtubes, or other nano- or micro-structured materials, into a foam structure that exhibits a negative Poisson's ratio. The transformation of a PPR foam or a nanostructured or microstructured material into an NPR foam can involve thermal treatment (e.g., heating, cooling, or both), application of pressure, or a combination thereof. In some examples, PPR materials, such as PPR foams or nanostructured or microstructured PPR materials, are transformed into NPR materials by chemical processes, e.g., by using glue. In some examples, NPR materials are fabricated using micromachining or lithographic techniques, e.g., by laser micromachining or lithographic patterning of thin layes of material. In some examples, NPR materials are fabricated by additive manufacturing (e.g., three-dimensional (3D) printing) techniques, such as stereolithography, selective laser sintering, or other appropriate additive manufacturing technique.

(32) In an example, a PPR thermoplastic foam, such as an elastomeric silicone film, can be

transformed into an NPR foam by compressing the PPR foam, heating the compressed foam to a temperature above its softening point, and cooling the compressed foam. Transformation processes can be done inside a predetermined dimension. In an example, a PPR foam composed of a ductile metal can be transformed into an NPR foam by uniaxially compressing the PPR foam until the foam yields, followed by uniaxially compression in other directions.

(33) NPR-PPR composite materials are composites that include both regions of NPR material and regions of PPR material. NPR-PPR composite materials can be laminar composites, matrix composites (e.g., metal matrix composites, polymer matrix composites, or ceramic matrix composites), particulate reinforced composites, fiber reinforced composites, or other types of composite materials. In some examples, the NPR material is the matrix phase of the composite and the PPR material is the reinforcement phase, e.g., the particulate phase or fiber phase. In some examples, the PPR material is the matrix phase of the composite and the NPR material is the reinforcement phase.

(34) NPR materials can exhibit various desirable properties, including high shear modulus, effective energy absorption, and high toughness (e.g., high resistance to indentation, high fracture toughness), among others.

(35) FIG. **4** shows a schematic depiction of the change in diameter of a material **400** upon impact. Although the material **400** in FIG. **4** is shown as a rounded ball, a similar deformation occurs in materials of other shapes. Prior to impact, the material **400** has a diameter d**1** in the direction of the impact and a diameter d**2** in the direction perpendicular to the impact. If the material **400** is a PPR material, the material undergoes significant deformation (e.g., elastic deformation) into a shape **402**, such that the diameter in the direction of the impact decreases to d**1**PPR and the diameter in the direction perpendicular to the impact increases to d**2**PPR. By contrast, if the material **400** is an NPR material, the material undergoes less extensive deformation into a shape **404**. The diameter of the shape **404** in the direction of the impact decreases to d**1**NPR, which is approximately the same as d**1**PPR. However, the diameter of the shape **404** in the direction perpendicular to the impact also decrease, to d**2**NPR. The magnitude of the difference between d**2** and d**2**NPR is less than the magnitude of the difference between d**2** and d**2**PPR, meaning that the NPR material undergoes less deformation than the PPR ball. This reduction in total deformation that is achievable by an NPR material enables the NPR material to have a different (e.g., longer) launching distance than an otherwise comparable PPR material at least in part because the NPR material has a lower wind resistance due to its smaller diameter upon compression.

(36) FIGS. **5**A and **5**B show plots of diameter versus time for a substantially spherical PPR material with a Poisson's ratio of 0.45 and an NPR material with a Poisson's ratio of −0.45, respectively, responsive to being struck with an equivalent force. In this example, the NPR material undergoes a smaller initial change in diameter than does the PPR material, and the oscillations in diameter are smaller in magnitude and dampen more quickly. Again, although FIGS. **5**A and **5**B are specific to substantially spherical materials, a similar behavior occurs in NPR and PPR materials of other shapes.

(37) FIG. **6** illustrates examples of NPR-PPR composite materials. An NPR-PPR composite material **602** is a laminar composite including alternating layers **604** of NPR material and layers **606** of PPR material. The layers **604**, **606** are arranged in parallel to a force to be exerted on the composite material **602**. Although the layers **604**, **606** are shown as having equal width, in some examples, a laminar composite can have layers of different widths.

(38) An NPR-PPR composite material **608** is a laminar composite including alternating layers of NPR material and PPR material, with the layers arranged perpendicular to a force to be exerted on the material **608**. In some examples, the layers of a laminar composite are arranged at an angle to the expected force that is neither perpendicular nor parallel.

(39) An NPR-PPR composite material **612** is a matrix composite including a matrix phase **611** of NPR material with a reinforcement phase **612** of PPR material. In the material **612**, the

reinforcement phase **612** includes fibers of the PPR material; in some examples, the reinforcement phase **612** can include particles or other configuration. In some examples, NPR-PPR composite materials can have a matrix phase of a PPR material with a reinforcement phase of an NPR material.

(40) FIG. **7** illustrates the mechanical behavior of PPR and NPR/PPR composite materials. A hypothetical block **700** of PPR material, when compressed along its width w, deforms into a shape **702**. The width w**1** of the compressed block **702** is less than the width w of the uncompressed block **700**, and the length l**1** of the compressed block **702** is greater than the length l of the uncompressed block: the material compresses along the axis to which the compressive force is applied and expands along a perpendicular axis.

(41) A block **704** of NPR/PPR composite material includes a region **708** of NPR material sandwiched between two regions **706** of PPR material. When the block **704** of composite material is compressed along its width, the material deforms into a shape **710**. The PPR regions **706** compress along the axis of compression and expand along a perpendicular axis, e.g., as described above for the block **700** of PPR material, such that, e.g., the width w**2** of a region **706** of uncompressed PPR material compresses to a smaller width w**4** and the length l**2** of the region **706** expands to a greater length l**4**. In contrast, the NPR region **708** compresses along both the axis of compression and along the perpendicular axis, such that, e.g., both the width w**3** and length l**3** of the uncompressed NPR region **708** are greater than the width w**5** and length l**5** of the compressed NPR region **708**.

(42) FIG. **8** illustrates an example method of making an object, such as an intraocular lens, formed of an NPR material. A granular or powdered material, such as a polymer material (e.g., a rubber) is mixed with a foaming agent to form a porous material **800**. The porous material **800** is placed into a mold **802**. Pressure is applied to compress the material **800** and the compressed material is heated to a temperature above its softening point. The material is then allowed to cool, resulting in an NPR foam **804**. The NPR foam **804** is covered with an outer layer **806**, such as a polymer layer, and heat and pressure is applied again to cure the final material into an object **808**.

(43) FIGS. **9**A-**9**F show embodiments of intraocular lenses that include a negative Poisson's ratio (NPR) material for the circumferential optic, each individual haptics, or any combination thereof. In the IOLs depicted in FIGS. **9**A-**9**F, the materials used to form the optical zone include a transparent polymer material having a positive Poisson's ratio (PPR) with a first index of refraction. The materials used to form the optical zone can include acrylic (e.g., polymethylmethacrylate), polycarbonate, silicone (e.g., silicone rubber), or hydrogel. One or more of the haptics, the circumferential optic, or all can be formed of a material having a negative Poisson's ration (NPR) with a second index of refraction different from the first index of refraction. For instance, the haptics, the circumferential optic, or combinations thereof can incorporate NPR materials of acrylics (e.g., PMMA), silicone, polycarbonate, or hydrogels. In the IOLs illustrated in FIGS. **9**A-**9**F, the surfaces of the optical zone can be convex or planar.

(44) FIG. **9**A shows a two-loop haptic embodiment **900** having two closed-loop haptics **902**. Each haptic **902** extends outwardly from the outer edge of a circumferential optic **906** of an optic **904**, and returns to be attached at the circumferential optic **906** to form a closed loop haptic structure. The two haptics **902** are at diametrically opposed positions around the circumference of the optic. Each haptic **902** is significantly smaller in width than the optic **904**. The circumferential optic **906**, one or more closed-loop haptics **902**, or all can be made of NPR material.

(45) FIG. **9**B shows an iris claw IOL **908**. The iris claw IOL **908** includes two oval haptics **910**, each of which is split in a middle **912** to form a pincer/claw-like structure. The two haptics **910** are attached to a circumferential optic **914** and disposed at diametrically opposed positions around a circumference of an optic **915**. One or more of the oval haptics **910**, the circumferential optic **914**, or all can be made of NPR material.

(46) FIG. **9**C shows a four plate-loop IOL **916** having two haptics **918**. Each haptic **918** has a width

approximately equal to the diameter of an optic **920**. The four plate-loop haptics are disposed at evenly spaced intervals around a circumference of the optic **920** and attached to the optic **920** at the circumferential optic **922**. One or more of the plate-loop haptics **918**, the circumferential optic **922**, or all can be made of NPR material.

(47) FIG. **9**D shows a solid plate IOL **924** having solid plate haptics **926**. Each solid plate haptic **926** has a width substantially equal to the diameter of an optic **928**. The two solid plate haptics **926** are disposed at diametrically opposed positions around a circumference of the optic **9288** and attached to the optic **928** at a circumferential optic **930**. One or more of the solid plate haptics **926**, the circumferential optic **930**, or all can be made of NPR material.

(48) FIG. **9**E shows a T-shaped haptic IOL **932** having T-shaped haptics **934**. The T-shaped haptics **934** are disposed at diametrically opposed positions around a circumference of an optic **936** and attached to the optic **936** at the circumferential optic **938** by respective hinges **934** that include grooves **942** in the haptic. One or more of the T-shaped haptics **934**, a circumferential optic **938**, or all can be formed of NPR material.

(49) FIG. **9**F shows an S-shaped haptic IOL **944** that includes two-loop haptics **946**. The two-loop haptics **946** are disposed at diametrically opposed positions around the circumference of an optic **948** and attached to the optic **948** at a circumferential optic **950**. Each two-loop haptics **946** includes an S-shape and has a width that is substantially less than a diameter of thean optic **948**. One or more of the haptics **946**, the circumferential optic **950**, or all can be formed of an NPR material.

(50) Other designs of the intraocular lens including different numbers and shapes of the haptics and radii of curvature of the optic surfaces are within the scope of this disclosure.

(51) FIG. **10** is a flow chart of an example process (**1000**) for implantation of an intraocular lens into an eye using surgical intervention. The surgery includes administering to the patient regional anesthesia, sedative anesthesia, or both (**1010**). A small incision is made in the sclera adjacent to the cornea or within the cornea (**1020**). The capsular bag is opened (**1030**). The natural lens may be removed from the eye (**1040**, **1050**) or it may be retained in the eye (**1035**). The IOL is inserted into the eye through the incision (**1045**, **1060**). The IOL may be positioned in place of the removed natural lens (**1070**), or it may be placed over the retained natural lens (**1055**), e.g., using one or more of the following techniques: insertion of a rigid IOL using a forceps; insertion of a folded, flexible IOL using a folding forceps, injection of a folded, flexible IOL using an injector loaded with the IOL, or another suitable technique. The IOL may be placed in the capsular bag, the anterior chamber, or the posterior chamber (**1070**). The IOL is disposed along an optical axis of the eye (**1080**) and one or more of the haptics are placed in contact with the tissue of the eye (**1090**). This tissue in contact with the haptics may include the capsular bag, the iris, the ciliary sulcus, the sclera, or a combination of these (**1090**). In some cases the IOL can be secured by positioning of the haptics without the need for further attachment. In some cases the haptics are further fastened using sutures, glue, or another suitable attachment mechanism, or a combination of these (**1090**). The incision in the sclera or the cornea is closed using fixation sutures, glue, cauterization, or another suitable closure mechanism, or a combination of these (**1095**).

(52) Other embodiments are within the scope of the following claims.

Claims

1. An intraocular lens comprising: a substantially circular lens element formed of a transparent material; and one or more haptics extending outwardly from an outer edge of the lens element, the entirety of each of the one or more haptics formed of a polymer foam material having a negative Poisson's ratio (NPR), in which the one or more haptics are configured to couple the intraocular lens to an eye of a patient.

2. The intraocular lens of claim 1, in which the lens element comprises: an inner region having a

first index of refraction; and an outer region disposed circumferentially surrounding the inner region, the outer region having a second index of refraction different from the first index of refraction.
3. The intraocular lens of claim 2, in which the outer region of the lens element is formed of a polymer foam NPR material.
4. The intraocular lens of claim 2, in which the inner region of the lens element is formed of a polymer material having a positive Poisson's ratio.
5. The intraocular lens of claim 1, in which an inner region of the lens element is formed of acrylic, silicone, or hydrogel.
6. The intraocular lens of claim 1, comprising two haptics disposed at diametrically opposed positions around a circumference of the lens element.
7. The intraocular lens of claim 1, comprising four haptics disposed at evenly spaced intervals around a circumference of the lens element.
8. The intraocular lens of claim 1, in which the one or more haptics each comprises a loop haptic.
9. The intraocular lens of claim 1, in which the one or more haptics each comprises a plate loop haptic.
10. The intraocular lens of claim 1, in which the one or more haptics each comprises a solid plate haptic.
11. The intraocular lens of claim 1, in which the one or more haptics each comprises a T-shaped haptic.
12. The intraocular lens of claim 1, in which a width of each of the haptics is less than a diameter of the lens element.
13. The intraocular lens of claim 1, in which a width of each of the haptics is substantially equal to a diameter of the lens element.
14. The intraocular lens of claim 1, in which each of the one or more haptics comprises a hinge attaching the haptic to the lens element.
15. The intraocular lens of claim 14, in which the hinge comprises grooves in the haptic.
16. The intraocular lens of claim 1, in which the polymer foam material is composed of a cellular structure having a characteristic dimension of between 0.1 μm and 3 μm.
17. The intraocular lens of claim 1, in which the polymer foam material comprises a foam of acrylic, silicone, or hydrogel.
18. The intraocular lens of claim 1, in which the polymer foam material comprises a re-entrant foam structure.
19. The intraocular lens of claim 1, in which the polymer foam material of the one or more haptics comprises a re-entrant foam structure.
20. The intraocular lens of claim 1, in which the polymer foam material of the one or more haptics is composed of a cellular structure having a characteristic dimension of between 0.1 μm and 3 μm.